SL ONLY - Due Feb 02nd

EXERCISE 1 [4 marks]

Prove that $(2n+5)^2-(2n-5)^2$ is a multiple of 10 for all $n \in \mathbb{Z}^+$.

EXERCISE 2

The first three terms of an arithmetic sequence are u_1, $5u_1-8$ and $3u_1+8$.

(a) Show that $u_1 = 4$. [2 marks]

(b) Prove that the sum of the first n terms of this arithmetic sequence is a square number. [2 marks]